

# MUSTAFA DOGAN

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🌐 dogan-mustafa    🔄 mustafaadogan

## EDUCATION

**Hacettepe University** 2021-2024  
Master's in Science in Computer Engineering | GPA: 3.64/4.0  
Ankara, Turkey  
Graduate Thesis: [Evaluating Zero-Shot Learning Capabilities of Vision-Language Models](#)  
Advisors: Erkut Erdem, Aykut Erdem

**Konya Food and Agriculture University (KFAU)** 2016-2021  
Bachelor's in Science in Computer Engineering | GPA: 3.72/4.0  
Konya, Turkey  
Graduation Project: *Autonomously Navigated Agent Using Reinforcement Learning*  
Advisor: Kasim Oztoprak

## ACADEMIC RESEARCH EXPERIENCE

**Fatima Fellowship** Sep 2024-Jun 2025  
Tarek Naous, Georgia Institute of Technology Remote  
– To Lie or Not to Lie? Investigating The Biased Spread of Global Lies by LLMs: [ACL'26 Main](#)

**Hacettepe University Computer Vision Lab** Feb 2022-Jun 2024  
Prof. Erkut Erdem, Prof. Aykut Erdem, Hacettepe University Ankara, Turkey  
– FewMMBench: A Benchmark for Multimodal Few-Shot Learning: [arXiv'26](#) 🧡 [mustafaa/FewMMBench](#)  
– Zero-Shot Linguistic Evaluation of Video Language Models: [ICLR'24](#)

**Network and Architecture Research Lab** Sep 2019- Jun 2021  
Prof. Kasim Oztoprak, Konya Food and Agriculture University Konya, Turkey  
– Teaching Computer Architecture from Bits and Bytes: [PeerJ Comp. Sci. Journal'24 \(Peer Reviewed\)](#)  
– Speculator and Influencer Analysis in Stock Market Using Social Media: [IEEE Big Data'20](#)

**Food Sensory Lab** Aug 2018  
Prof. Sungeun Cho, Michigan State University MI, USA  
– Project: Exploring Sleep-Eating Habits Using Python and Bayesian Analysis for Industry Insights

## INDUSTRY RESEARCH EXPERIENCE

**Aselsan** Nov 2021 - Present  
Mid-Level Software Engineer Ankara, Turkey  
◦ Developed end-to-end firing control software for multiple air-defense platforms in modern C++, integrating distributed subsystems via Serial, Ethernet, CAN, and DDS.  
◦ Developed on-premises LLM-based developer tools for code review, software assistance, and knowledge retrieval, leveraging RAG and agentic workflows.

**Morten Inc.** Jun - Sept 2020  
Machine Learning Engineer Intern Istanbul, Turkey  
◦ Created a *Knowledge Graph* to catalog authors and citations using *GPT-2*, *Python*, and *MongoDB*.  
◦ Built a *GPT-2* to application to identify future tasks in academic papers, aiding in paper clustering.

## PUBLICATIONS & PRE-PRINTS

ACL Main 2026 [To Lie or Not to Lie? Investigating The Biased Spread of Global Lies by LLMs](#)  
Z Khan\*, **M Dogan\***, I Okoh, P Sadeghi, S Shrestha, SJ Nyah, MO Mokhiamar, MJ Ryan, T Naous

arXiv 2026 [FewMMBench: A Benchmark for Multimodal Few-Shot Learning](#)  
**M Dogan**, I Kesen, I Calixto, A Erdem, E Erdem

ICLR 2024 [ViLMA: A Zero-Shot Benchmark for Linguistic and Temporal Grounding in Video-Language Models](#)  
I Kesen, A Pedrotti, **M Dogan**, M Cafagna, EC Acikgoz, L Parcalabescu, I Calixto, A Frank, A Gatt, A Erdem, E Erdem

PeerJ Comp. Sci. [Teaching computer architecture by designing and simulating processors from their bits and bytes](#)  
2024 **M Dogan**, K Oztoprak, MR Tolun

IEEE Big Data [Speculator and Influencer Evaluation in Stock Market by Using Social Media](#)  
2020 **M Dogan**, O Metin, E Tek, S Yumusak, K Oztoprak

## OPEN SOURCE SOFTWARE/DATA

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**GlobalLies**  [zohaib-khan5040 / globallies](#)

GlobalLies is a multilingual, parallel dataset of misinformation prompts and entities designed to systematically evaluate how LLMs generate and propagate misinformation across different languages and countries.

**FewMMBench**  [mustafaa/FewMMBench](#)

FewMMBench is a benchmark designed to evaluate the few-shot learning capabilities of MLLMs under ICL and CoT prompting across a diverse set of multimodal understanding tasks involving image-text inputs.

**ViLMA Benchmark**  [ilkerkesen/ViLMA](#)

ViLMA benchmarks Video-Language Models in linguistic and temporal grounding across five dimensions: action counting, situation awareness, state change, rare actions, and spatial relations.

**DoMe Architecture Simulator**  [omer-metin/CPU16-Simulator](#)  [mustafaadogan/RISC16-Verilog](#)  [YouTube Playlist](#)

The DoMe architecture features a custom instruction set and data-path with 18 essential instructions for memory and register operations. A user-friendly desktop simulator, built in Python with PyQt, enables execution of assembly programs while displaying real-time memory and register values. The design and functionality were verified through a Verilog implementation in the ISE development environment.

**Stock Market Analysis Datasets**  [Tweet Dataset](#)  [NASDAQ Value Dataset](#)  [omer-metin/TweetCollector](#)

This project utilizes two datasets: one consisting of 3.4 million NASDAQ-related tweets, which helps identify market speculators and influencers by collecting tweet content, user information, and engagement metrics through Selenium (Tweet-Collector) and storing them in an SQLite database, and another containing daily market share value changes for Amazon, Apple, Google, Microsoft, and Tesla from mid-2010 to mid-2020.

## VOLUNTEERING ACTIVITIES

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**Erasmus Youth Exchange Project, *Interstellar Inclusion***

Oct 2019

*Participant*

*Perugia, Italy*

- Actively engaged in fostering understanding and combat *racism* and *xenophobia* by peer experiences.
- Collaborated in a theater performance with a diverse group to address educational access challenges.
- Created a short movie on Freedom, winning the 'Best Movie' accolade among five nominees.  [Movie](#)
- Delivered a presentation on migration in Turkey to 30+ youngsters from diverse backgrounds.

**Erasmus Youth Exchange Project, *Sail Entrepreneurship***

May 2018

*Participant*

*Nelijärve, Estonia*

- Fostered open communication and encouragement among a group of 35+ participants.
- Developed business concepts focusing on social entrepreneurship using the *Business Canvas Model*.

## AWARDS AND ACHIEVEMENTS

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- 2024 **Granted of 9-month program; top 10% of 500+ applicants for technical skills**, Fatima Fellowship
- 2023 **Achievement Award**, Aselsan
- 2020 **Best Course Project Award**, CENG3204 Artificial Intelligence
- 2019 **Best Movie Award**, Erasmus+ Project (Interstellar Inclusion)
- 2018 **Awarded \$6,000 for a summer internship grant at Michigan State University**, KFAU
- 2016-21 **Dean's Honor Award**, KFAU
- 2<sup>nd</sup> highest rank in the faculty**, KFAU
- Recipient of merit-based full scholarship and monthly stipend by Dean's Office**, KFAU